



Brief paper

Redundant measurement-based second order mutual difference adaptive Kalman filter[☆]

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ABSTRACT

Noise distribution plays an essential role in state estimation using Kalman filter. However, statistical characteristics of the noise are often unknown in most practical applications. A second order mutual difference (SOMD) algorithm has been proposed to generate an estimation of the measurement noise covariance matrix R by calculating the autocorrelation of SOMD of redundant measurements, and thus it can avoid coupling with the state estimation error; however, the algorithm cannot be applied directly for a majority of practical systems due to the requirement of redundant measurements. In this paper, the SOMD algorithm is expanded to the system with single measurement by constructing a pseudo measurement. A non-zero estimation bias detection algorithm is presented to address the inconsistency between the mathematical model and the real. A modified robust adaptive Kalman filter (RAKF) is also developed to tackle this inconsistency and improve filtering accuracy by activating adaptive operation properly. The efficacy of the approach is demonstrated via a target tracking problem. Simulation results indicate that the proposed algorithm can reflect the noise properties accurately and outperform several reference algorithms in precision and robustness.

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1. Introduction

In addition to appropriate models, the key to achieving better estimation using the Kalman filter lies in effective extraction of the available process and measurement information (Kalman, 1960). For this reason, statistical properties of noise are crucial for state estimation; however, a priori knowledge of noise is often unknown or potentially time-varying in most actual applications. The use of inaccurate a priori statistics will lead to precision degradation and even divergence of the estimated state when the traditional Kalman filter is employed (Mehra, 1972). To address this problem, considerable efforts have been made to modify the stochastic information in traditional Kalman filter and identify the correct noise properties, leading to the development of adaptive Kalman filter (e.g., Jo, Chu, & Sunwoo, 2012; Karasalo & Hu, 2011; Li & Bar-Shalom, 1994; Myers & Tapley, 1976; Petersen & Koch, 2010).

The most popular adaptive Kalman filtering approaches can be found in the literature are multiple model adaptive estimation (MMAE), innovation-based adaptive estimation (IAE) and residual-based adaptive estimation (RAE).

MMAE was first presented by Magill (1965) to estimate states when the model for describing system dynamics is uncertain. It is helpful in applications such as fault detection and when the system model changes with some unknown events (Gadsden, Song, & Habibi, 2013; Kim, Choi, & Kim, 2008). However, as a precondition for utilizing the method is that one model in the candidate bank is the real one, MMAE is inappropriate for the system with unknown dynamics (Li & Bar-Shalom, 1996; Li & Jilkov, 2005).

As extensions to the traditional Kalman filtering, the IAE and RAE algorithms use extra procedures to estimate the process noise covariance matrix Q and/or the measurement noise covariance matrix R , where Q and R are adapted as measurements evolve with time. A detailed description of the two schemes can be found in Mohamed and Schwarz (1999). More recent advances have been achieved on this topic. Odelson, Rajamani, and Rawlings (2006) proposed an autocovariance least-squares approach to estimate Q and R by using the autocovariance function of the innovation and its lagged version. Robust adaptive Kalman filter (RAKF) proposed by Yang and Gao (2006) utilizes an adaptive factor to tune the predicted state error covariance P^- . The adaptive factor is computed based on the relationship between the covariance matrix

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of the innovation vector and its corresponding theoretical one. The value of the factor is obtained to be greater than 1 so that RAKF can provide heavier weight to new measurement and thus enhance the robustness of the filter. Similarly, the adaptive scaling algorithm developed by [Ding, Wang, Rizos, and Kinlyside \(2007\)](#) adjusts the covariance of process noise by applying a scaling factor, which can be either greater or smaller than 1, thus giving a full range of options to tune Q . However, all the algorithms described above are based on the information provided by the innovation, which is the difference between the new measurement and its prediction; or the residual, which is the difference between the new measurement and its updated value from the Kalman filter. Accordingly, the state estimation error can directly corrupt the estimation of noise covariance or state error covariance matrix, and thereby lead to inaccurate and even divergent output of the filter ([Fitzgerald, 1971](#)).

Recent attempts for nonlinear estimation include recursive update filter (RUF) ([Zanetti, 2012](#)) and adaptable recursive update filter (ARUF) ([Zanetti, 2015](#)). The RUF utilizes the update gradually and re-linearizes each recursion. However, the number of recursion in each step is constant and user-defined; therefore, the method can be unsatisfactory when the estimated time is of concern. To reduce unnecessary recursions, the ARUF is developed to adaptively select the number of recursion. Nonetheless, the two methods cannot tune the measurement noise statistics according to different circumstances.

To mitigate these defects, a second order mutual difference (SOMD) algorithm is proposed to estimate R based on redundant measurements ([Zhang, Chang, & Che, 2010](#)). The main idea of SOMD is to utilize one measurement as a reference to evaluate the noise covariance of the other; therefore, the estimation of R can be absolutely decoupled from the state estimation error. However, the algorithm cannot be applied to a majority of practical systems because of the requirement for redundant measurements, which is often unsatisfied in consideration of cost.

This paper proposes a redundant measurement-based adaptive Kalman filtering (RMAKF) algorithm. The algorithm estimates R by utilizing the SOMD algorithm in the system with single measurement by establishing a pseudo measurement. Furthermore, a rule for detecting the non-zero estimation bias is presented, and the RAKF is used for compensation when the system suffers from estimation bias.

The main contribution of this paper can be summarized as follows: (1) The proposed algorithm extends the SOMD algorithm to a system with single measurement by constructing a pseudo measurement, and make the R estimation algorithm applicable to a wide range of applications. (2) We combine the R estimation algorithm with RAKF approach to improve the accuracy of state estimation. A rule for detecting the non-zero estimation bias is also investigated as a trigger condition of RAKF.

The remainder of this paper is organized as follows: Section 2 introduces the related work on some existing adaptive Kalman filters. Section 3 presents the R estimation algorithm based on pseudo SOMD (PSOMD). An adaptive filtering algorithm based on the R estimation method, as well as a rule for detecting estimation bias, is provided in Section 4. Section 5 gives a numerical simulation of a target tracking example and discusses the simulation results. Finally, a conclusion is drawn in Section 6.

Notations: $\mathbb{R}$ denotes the set of all real numbers, $\mathbb{R}^k$ k -dimensional Euclidean space with 2-norm $\|\cdot\|$, and $\mathbb{R}^{m \times n}$ is the vector space of all $m \times n$ matrices with entries in $\mathbb{R}$. I and 0 refer to the identity and zero matrices of appropriate dimensions, respectively. A^T represents the transpose of matrix A ; $\text{tr}(A)$ stands for the trace of matrix A . Finally, $E(\cdot)$ is the mathematical expectation; $\text{Var}(\cdot)$ is the abbreviation of variance.

2. Related work

Consider a stochastic linear discrete-time system

$$\begin{cases} X_k = \Phi_{k,k-1}X_{k-1} + W_{k-1}, \\ Z_k = H_kX_k + V_k, \end{cases} \quad (1)$$

where $X_k \in \mathbb{R}^n$, $Z_k \in \mathbb{R}^m$, $\Phi_{k,k-1} \in \mathbb{R}^{n \times n}$ and $H_k \in \mathbb{R}^{m \times n}$. Note that $W_k \in \mathbb{R}^p$ and $V_k \in \mathbb{R}^m$ are independent random noise vectors with covariance matrices $Q_k = E(W_k W_k^T)$ and $R_k = E(V_k V_k^T)$, respectively. Additionally, $E(V_k V_j^T) = R_k \delta_{k-j}$, $k, j \in \{1, 2, \dots, N\}$, where δ_{k-j} is the Kronecker delta function, i.e., $\delta_{k-j} = 1$ if $k = j$ and $\delta_{k-j} = 0$ if $k \neq j$.

The estimation of measurement noise properties is typically crucial to adaptive Kalman filter, but some existing adaptive algorithms, such as IAE/RAE, RAKF and ARUF, have inherent deficiency in this regard.

Based on the assumption of white noise, the noise covariances in the IAE and RAE are adapted as follows:

$$\begin{cases} \hat{R}_k^{\text{iae}} = E[v_k^-(v_k^-)^T] - H_k P_k^- H_k^T, \\ \hat{R}_k^{\text{rae}} = E[v_k^+(v_k^+)^T] + H_k P_k H_k^T, \end{cases} \quad (2)$$

where $v_k^- = Z_k - H_k \hat{X}_k^-$ and $v_k^+ = Z_k - H_k \hat{X}_k^+$ are the innovation and residue at time k , respectively; $\hat{X}_k^-$ represents the a priori estimated state; $\hat{X}_k^+$ represents the updated filter state; P_k^- denotes the covariance of the estimation error of $\hat{X}_k^-$; P_k denotes the covariance of the estimation error of $\hat{X}_k^+$. It is obvious that the innovation and residual are sensitive to $\hat{X}_k^-$ and $\hat{X}_k^+$, respectively. As a result, the inaccuracy of $\hat{X}_k^-$ or $\hat{X}_k^+$ will affect the estimation of R .

In RAKF, an adaptive factor is introduced to depress the estimation error by expanding estimation error covariance P_k^- . The adjustment algorithm is

$$K_k = \frac{1}{\alpha_k} P_k^- H_k^T (\frac{1}{\alpha_k} H_k P_k^- H_k^T + R_k)^{-1}, \quad (3)$$

where K_k is the Kalman gain matrix, and $\alpha_k (0 < \alpha_k \leq 1)$ is the adaptive factor and can be expressed as

$$\begin{cases} \alpha_k = \begin{cases} 1, & |\Delta v_k^-| \leq c \\ \frac{c}{|\Delta v_k^-|}, & |\Delta v_k^-| > c \end{cases} \\ \Delta v_k^- = \left[\frac{(v_k^-)^T v_k^-}{\text{tr}(\Sigma_k^-)} \right]^{\frac{1}{2}}, \\ \Sigma_k^- = H_k P_k^- H_k^T + R_k, \end{cases} \quad (4)$$

where c is a presupposed threshold, and is generally set to $1 \sim 2.5$ ([Yang & Gao, 2006](#)); Δv_k^- is discriminating statistical information based on the innovations; Σ_k^- is the covariance matrix of the innovation vector v_k^- . As shown in (3), the RAKF magnifies the matrix P_k^- by introducing the adaptive factor α_k , and then enlarges the Kalman gain K_k . This means more dependence upon the new measurements. Consequently, the performance of the filter will degrade when the assumed R violates its real value. Additionally, the algorithm cannot discriminate measurement error from estimation error, this can lead to large estimation error.

The ARUF applies the update gradually and re-linearizes each recursion to track the nonlinearity of the measurement exactly. The algorithm takes advantage of the relationship between the residual ratio and the innovation ratio to keep each recursion in the region where the linearization assumption holds. In the presence of nonlinear measurements, it can be derived that

$$(v_k^+)^T (\Sigma_k^+)^{-1} v_k^+ = (v_k^-)^T (\Sigma_k^-)^{-1} v_k^- + \kappa, \quad (5)$$

where Σ_k^+ is the covariance matrix of the residual vector v_k^+ , and κ is some scalar function of the higher-order terms. Only when

the measurement is linear, the residual ratio equals the innovation ratio in (5). This property is used to adaptively select the number of recursion N . More specifically, if the residual ratio does not match the innovation ratio within a threshold ϑ , increase N by 1. Although the method is susceptible to the measurement noise, it cannot tune R . If the underlying assumptions of R do not hold, the method will attribute the measurement error to the estimation error and make more recursions, and then generate inaccurate estimation.

To mitigate these defects, SOMD algorithm is proposed to estimate R based on redundant measurements, and the estimated R is decoupled from the state estimation error. The SOMD algorithm is as follows.

Suppose that Z_k^1 and Z_k^2 are the redundant measurements of Z_k , obtained from different instruments, with uncorrelated Gaussian zero-mean white noises V_k^1 and V_k^2 (with respective variances R_k^1 and R_k^2) at time k , respectively. If one noise covariance can be neglected in comparison with the other one as follows:

$$E[V_k^2(V_k^2)^T] \ll E[V_k^1(V_k^1)^T], \quad (6)$$

then the greater covariance can be obtained by

$$\hat{R}_k^1 \approx \text{Var}(\Delta Z_k^1 - \Delta Z_k^2)/2, \quad (7)$$

where $\Delta Z_k^i = Z_k^i - Z_{k-1}^i$ ($i = 1, 2$) is the first order self-difference sequence of each measurement, and $\Delta Z_k^1 - \Delta Z_k^2$ is the SOMD sequence.

The estimation of R does not depend on the estimated state, and is immune to the process noise. It has been proven that the SOMD algorithm improves the performance of INS/GPS integrated navigation system, which has typical V_k^2 in INS and V_k^1 in GPS (Zhou, Zhang, Li, & Li, 2015). However, the algorithm cannot be applied to a majority of practical systems because of the requirement of redundant measurements and difficulty in satisfying condition (6). For that reason we will present an approach to constructing a pseudo measurement in the next section.

3. Estimation of R based on PSOMD

To extend the SOMD algorithm to a system without redundant measurements, some measurement references must be provided. In this section, we present a method for constructing a pseudo measurement, in which the noise must be a zero-mean white noise and uncorrelated with the direct measurement noise.

The estimation of state can be intuitively utilized to generate the reference measurement, such as $H_k \hat{X}_k$. However, it is disqualified because $\hat{X}_k$ is the expectation of X_k conditioned on all the measurements up to and including time k and thus the noise in $\hat{X}_k$ is correlated with measurement noise V_k . In the following theorem, we will introduce a pseudo measurement and estimate R based on PSOMD (R-PSOMD).

Theorem 1. For a given system with zero-mean process and measurement white noises as described by (1), the adjacent pseudo measurements Z_k^p , Z_{k-1}^p and their first order self-difference ΔZ_k^p are defined as

$$\begin{cases} Z_k^p = H_k \hat{X}_{k|k-2}, \\ Z_{k-1}^p = H_{k-1} \hat{X}_{k-1|k-2}, \\ \Delta Z_k^p = Z_k^p - Z_{k-1}^p. \end{cases} \quad (8)$$

If the sampling time T_s is small, then the covariance of measurement noise can be estimated as (9) when the Kalman filter is stable.

$$\hat{R}_k \approx [\text{Var}(\Delta Z_k - \Delta Z_k^p) - H_k Q_{k-1} H_k^T]/2. \quad (9)$$

Proof. According to system (1), the first order self-difference sequence of direct measurement is

$$\begin{aligned} \Delta Z_k &= Z_k - Z_{k-1} \\ &= (H_k X_k + V_k) - (H_{k-1} X_{k-1} + V_{k-1}) \\ &= (Z_k^r - Z_{k-1}^r) + (V_k - V_{k-1}), \end{aligned} \quad (10)$$

where Z_k^r and Z_{k-1}^r , defined by the above equation, are the ideal measurements of Z_k and Z_{k-1} , respectively.

Considering (1), the two-step and one-step predictions based on the same filtered state are

$$\begin{cases} \hat{X}_{k|k-2} = \Phi_{k,k-2} \hat{X}_{k-2}, \\ \hat{X}_{k-1|k-2} = \Phi_{k-1,k-2} \hat{X}_{k-2}. \end{cases} \quad (11)$$

Denoting the error between the true state and the estimated state as $\tilde{X}$, i.e.,

$$\tilde{X}_{k-2} = X_{k-2} - \hat{X}_{k-2}, \quad (12)$$

and then the first order self-difference of Z_k^p is

$$\begin{aligned} \Delta Z_k^p &= Z_k^p - Z_{k-1}^p \\ &= H_k \hat{X}_{k|k-2} - H_{k-1} \hat{X}_{k-1|k-2} \\ &= H_k \Phi_{k,k-2} \hat{X}_{k-2} - H_{k-1} \Phi_{k-1,k-2} \hat{X}_{k-2} \\ &= H_k \Phi_{k,k-2} (X_{k-2} - \tilde{X}_{k-2}) \\ &\quad - H_{k-1} \Phi_{k-1,k-2} (X_{k-2} - \tilde{X}_{k-2}) \\ &= H_k \Phi_{k,k-2} X_{k-2} - H_{k-1} \Phi_{k-1,k-2} X_{k-2} \\ &\quad - (H_k \Phi_{k,k-2} - H_{k-1} \Phi_{k-1,k-2}) \tilde{X}_{k-2} \\ &= H_k (X_k - \Phi_{k,k-1} W_{k-2} - W_{k-1}) \\ &\quad - H_{k-1} (X_{k-1} - W_{k-2}) \\ &\quad - (H_k \Phi_{k,k-2} - H_{k-1} \Phi_{k-1,k-2}) \tilde{X}_{k-2} \\ &= (Z_k^r - Z_{k-1}^r) - V_k^p, \end{aligned} \quad (13)$$

where

$$\begin{aligned} V_k^p &= H_k W_{k-1} + (H_k \Phi_{k,k-1} - H_{k-1}) W_{k-2} \\ &\quad + (H_k \Phi_{k,k-2} - H_{k-1} \Phi_{k-1,k-2}) \tilde{X}_{k-2}. \end{aligned} \quad (14)$$

Considering (10), the SOMD can be obtained as

$$\Delta Z_k - \Delta Z_k^p = (V_k - V_{k-1}) + V_k^p. \quad (15)$$

Note that $\tilde{X}_{k-2}$, including the information of zero-mean white noises $(W_1, \dots, W_{k-3}, V_1, \dots, V_{k-2})$, terminates at time $k-2$. Since V_k^p is a linear combination of X_{k-2} , W_{k-1} and W_{k-2} , the two items on the right hand side of (15) are uncorrelated. Also, the noise statistical characteristic remains the same at adjacent steps. Therefore, taking the variance of (15) gives

$$\text{Var}(\Delta Z_k - \Delta Z_k^p) = 2R_k + \text{Var}(V_k^p). \quad (16)$$

Then, we have

$$R_k = [\text{Var}(\Delta Z_k - \Delta Z_k^p) - \text{Var}(V_k^p)]/2. \quad (17)$$

For a time-invariant system, $H_k = H_{k-1}$. In general, $H_k \approx H_{k-1}$ in a short time, then (14) becomes

$$\begin{aligned} V_k^p &\approx H_k W_{k-1} + H_k (\Phi_{k,k-1} - I) W_{k-2} \\ &\quad + H_k (\Phi_{k,k-2} - I) \Phi_{k-1,k-2} \tilde{X}_{k-2}. \end{aligned} \quad (18)$$

Taking the variance of V_k^p results in

$$\begin{aligned} \text{Var}(V_k^p) &= H_k Q_{k-1} H_k^T + H_k (\Phi_{k,k-1} - I) Q_{k-2} (\Phi_{k,k-1} - I)^T H_k^T \\ &\quad + [H_k (\Phi_{k,k-1} - I) \Phi_{k-1,k-2}] P_{k-2} \\ &\quad \times [H_k (\Phi_{k,k-1} - I) \Phi_{k-1,k-2}]^T. \end{aligned} \quad (19)$$

If $\Phi_{k,k-1} = I$, $\text{Var}(V_k^p) = H_k Q_{k-1} H_k^T$ is absolutely true. Moreover, if A is the system matrix of the corresponding continuous-time system, the state transition matrix can be written as $\Phi_{k,k-1} = e^{AT_s}$. For a small T_s , $e^{AT_s} \approx I$, then $(\Phi_{k,k-1} - I) \rightarrow 0$. Thus, (19) can be simplified as

$$\text{Var}(V_k^p) \approx H_k Q_{k-1} H_k^T. \quad (20)$$

Substituting (20) into (17), we can obtain (9). The proof is accomplished. $\square$

Remark 1. Recalling the IAE algorithm in (2), considering the definition of innovation, $E[v_k^-(v_k^-)^T]$ can be seen as the variance of the first order difference of measurement. The higher the difference order, the greater the variance. Then $E[v_k^-(v_k^-)^T] < \text{Var}(\Delta Z_k - \Delta Z_k^p)$. Also, because $P_k^- = \Phi_{k,k-1} P_{k-1}^+ \Phi_{k,k-1}^T + Q_{k-1}$, then $H_k P_k^- H_k^T > H_k Q_{k-1} H_k^T$. Comparing (2) with (9), we have $\text{Var}(\Delta Z_k - \Delta Z_k^p) - H_k Q_{k-1} H_k^T > E[v_k^-(v_k^-)^T] - H_k P_k^- H_k^T$. Therefore, the proposed algorithm is less prone to generate a non-positive definite matrix than IAE.

Theorem 1 constructs a pseudo measurement which can avoid coupling with the noise of direct measurement. It also proposes the PSOMD to estimate R , in which the state estimation error can be neglected when T_s is small. In addition, the estimation stability of the proposed algorithm is better than that of IAE.

4. Adaptive filtering based on R-PSOMD

As discussed above, estimation of R based on PSOMD can be employed to improve the performance of Kalman filter. However, only adaptive R is inadequate to cope with complex situations. In practice, it is common that a mathematical model may be inconsistent with reality, while RAKF is a strong tracking method which can compensate for the estimation bias caused by modeling errors. In order to improve the precision and robustness, we propose an adaptive filtering scheme which combines R-PSOMD with RAKF. The main idea of the adaptive filter is to detect the modeling error precisely, and then switch on/off RAKF.

4.1. Triggering condition of RAKF

A window based mean of the difference between the estimation and prediction of each state x^i ($i = 1, 2, \dots, n$) is written as

$$M(i, k, w) = \frac{1}{w} \sum_{j=0}^{w-1} (\hat{x}_{k-j}^i - \hat{x}_{k-j|k-j-2}^i), \quad (21)$$

where w represents the window length, and is generally set to 30 ~ 50.

We fit a straight line based on $M(i, k, w)$ over successive $s + 1$ windows, and let the line be

$$f(\lambda) = \Theta_1 + \Theta_2(k - \lambda w). \quad (22)$$

Then the intercept and slope of (22), $[\Theta_1 \ \Theta_2]^T$, can be recognized with a least squares (LS) sense by minimizing

$$\sum_{\lambda=0}^s [M(i, k - \lambda w, w) - f(\lambda)]^2. \quad (23)$$

Introducing the notations

$$\Lambda = \begin{bmatrix} 1 & k \\ 1 & k - w \\ \vdots & \vdots \\ 1 & k - sw \end{bmatrix}, \quad \Xi = \begin{bmatrix} M(i, k, w) \\ M(i, k - w, w) \\ \vdots \\ M(i, k - sw, w) \end{bmatrix},$$

the solution for the LS estimation is

$$[\Theta_1 \ \Theta_2]^T = (\Lambda^T \Lambda)^{-1} \Lambda^T \Xi. \quad (24)$$

If there exists an estimation bias, the window based mean in (21) will depart from zero, or the line (22) will exhibit an increasing or decreasing trend which is statistically distinguished from random behaviors. This property is used as the triggering condition of RAKF.

Then the method for bias detection can be made as below. If one of the following conditions holds, RAKF is started.

$$\begin{cases} |M(i, k, w)| \geq \delta_i, \\ |\Theta_2| \geq \varepsilon, \end{cases} \quad (25)$$

where δ_i denotes the bias threshold of state x^i , and is a very small positive number near zero, and $\varepsilon > 0$ is the threshold of the slope.

If there is a sufficient bias of some main states which can affect the result much more, RAKF will be activated to compensate for modeling errors.

4.2. Computational implementation of R-PSOMD

In practice, in order to adapt to the variation of R in time, the width of a sliding window should be limited, and this is easy to cause the error of R-PSOMD due to insufficient samples. To suppress this adverse impact, a fading memory calculation is implemented as below:

$$\hat{R}_k = (1 - d_k) \hat{R}_{k-1} + d_k \hat{R}_k^-, \quad (26)$$

with

$$\begin{aligned} d_k &= (1 - b)/(1 - b^{k+1}), \\ \hat{R}_k^- &= \frac{1}{2w_R} \sum_{i=0}^{w_R-1} (\Delta Z_{k-i} - \Delta Z_{k-i}^p)(\Delta Z_{k-i} - \Delta Z_{k-i}^p)^T \\ &\quad - \frac{1}{2} H_k Q_{k-1} H_k^T, \end{aligned}$$

where b is a forgetting factor and is selected as 0.995 in our work, w_R is the sliding window width, $\hat{R}_k^-$ denotes the direct output at time k , and $\hat{R}_k$ is the final covariance matrix. Moreover, the estimation effect of R is closely connected to the sampling frequency, the higher it is, the better the estimation we can obtain.

5. Simulations

In this section, a numerical example of a radar tracking problem is given to illustrate the effectiveness of the proposed adaptive filtering approach.

Consider a target trajectory with linear motion and maneuver motion in 3D Cartesian coordinates, as shown in Fig. 1. The tracking time is 140 s. The target moves in straight line with a constant velocity of 200 m/s for the first 50 s, followed by a uniform circular motion with the same speed and cycle of 100 s from 50 s to 100 s, and finally flies back along a straight line with a velocity of 200 m/s.

5.1. Model for target tracking problem

The estimated state consists of positions in three spatial directions and the velocity toward those coordinate axes. The radar station, located at the origin of reference coordinate, is used to measure distance, azimuth and pitch angles of the target at a sampling period of 0.1 s. Thus, the model for target tracking problem can be represented by

$$\begin{cases} \dot{X}(t) = FX(t) + W(t), \\ Z_k = h(X_k) + V_k, \end{cases} \quad (27)$$

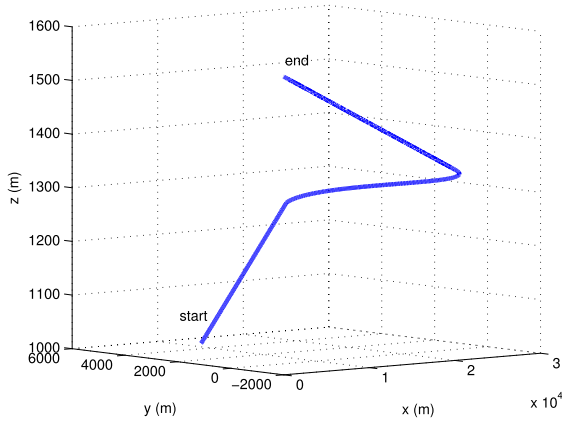


Fig. 1. Target trajectory.

where

$$F = \begin{bmatrix} 0_{3 \times 3} & I_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix}$$

$$h(X) = \begin{bmatrix} r \\ \beta \\ \alpha \end{bmatrix} = \begin{bmatrix} \sqrt{x_1^2 + x_2^2 + x_3^2} \\ \tan^{-1}(x_1/x_2) \\ \tan^{-1}(x_3/\sqrt{x_1^2 + x_2^2}) \end{bmatrix},$$

$W(t)$ is the zero-mean process noise vector, and its power spectral density is

$$Q = \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & 10^{-5} I_{3 \times 3} \end{bmatrix},$$

r represents the range, and β and α stand for the azimuth and pitch angles, respectively. $V_k \sim N(0, R_k)$, is the measurement noise at epoch k , and is independent of $W(t)$. To evaluate the proposed method, we set the real measurement noise covariance matrix

according to the following rules:

$$R_k = \begin{cases} \text{diag}(2^2, (0.02\pi/180)^2, (0.02\pi/180)^2), & 0 \leq t \leq 100 \text{ s}, \\ \text{diag}(10^2, (0.06\pi/180)^2, (0.06\pi/180)^2), & 100 \text{ s} < t \leq 140 \text{ s}. \end{cases}$$

5.2. Discussion of simulation results

As stated before, the tracking problem is nonlinear, while the assumption behind the R-PSOMD is that the measurement system is linear. Therefore, we capitalize on EKF to estimate the state. By comparing the simulation results of RAKF, RAE and ARUF, the performance of the proposed method is evaluated.

Fig. 2 exhibits the estimations of measurement noise covariance provided by RAE and PSOMD. As depicted on the top of the figure, the estimations given by RAE deviate from the true value seriously during the maneuvering motion. This can be ascribed to its reliance on the residual and accumulation of state estimation error. However, PSOMD can contribute accurate and stable estimations over almost the whole process except the transition phase after the sudden change, which is caused by the lower sampling frequency. We increase the sampling frequency to 100 Hz and make a comparison again, the corresponding performance is demonstrated at the bottom of Fig. 2. Again, PSOMD gives more accurate and stable estimations than RAE.

The estimated position errors of all the algorithms are depicted in Fig. 3. The performance of the proposed algorithm with no adaptation of R (NAR) is also displayed for the sake of comparison. It shows that all the algorithms give almost identical results before 50 s, but behave differently later. RAE gives the worst performance during the maneuvering motion, because it works based on the residual sequence and is coupled with state estimation error. In addition, RAKF yields oscillatory estimations while experiencing the lower half of linear motion where the measurement noise is larger. This point is also shown in Fig. 4, which depicts the estimated velocity errors of all the algorithms. Nevertheless, the proposed algorithm gives more accurate and robust estimation

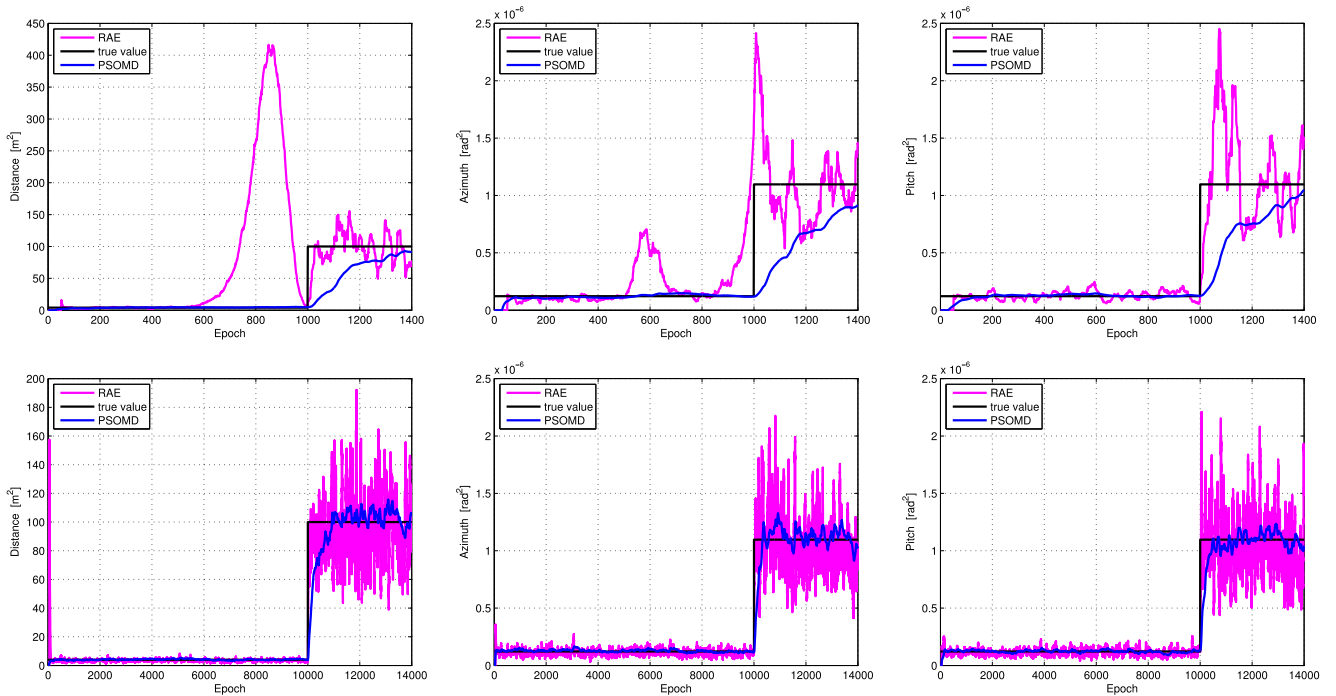


Fig. 2. Estimation of R . Top: $T_s = 0.1$ s. Bottom: $T_s = 0.01$ s. Left: distance. Middle: azimuth. Right: pitch.

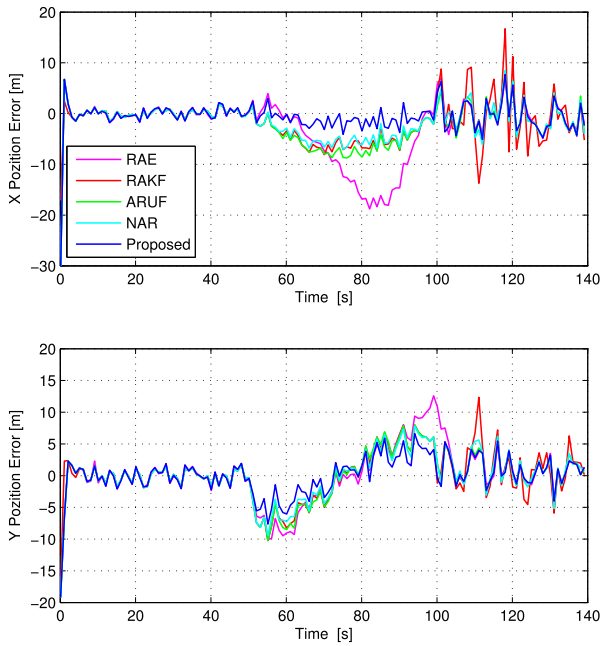


Fig. 3. Estimated position errors in X and Y axes.

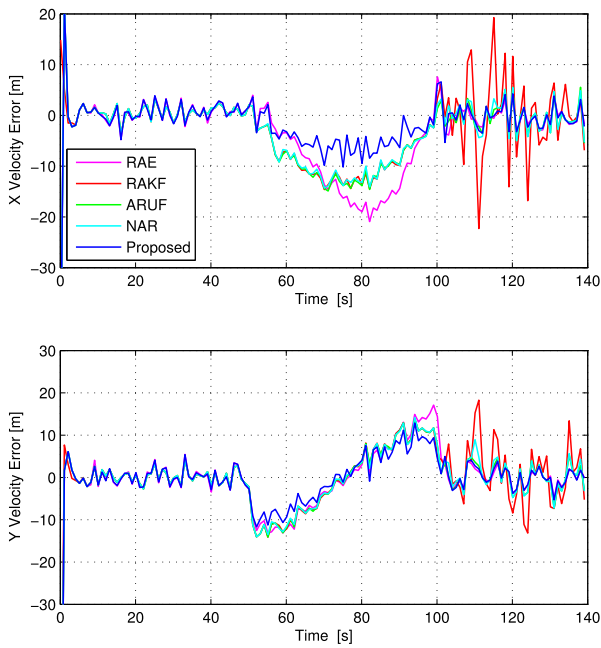


Fig. 4. Estimated velocity errors in X and Y axes.

than other algorithms. The estimation errors of NAR also indicate the estimation of R is efficient.

Fig. 5 shows the starting time of RAKF in the proposed algorithm. We set $\delta = 0.5$ and $\varepsilon = 0.5$ in the bias detection rule in (25), which is also the triggering condition of RAKF. The starting time of RAKF is concentrated in the maneuvering motion where the model is inaccurate, which indicates the bias detection in Section 4.1 is efficient.

In this numerical example, we set $\vartheta = 0.02$ in accordance with the theory in ARUF. The number of recursions selected by the algorithm varies between 2 and 3 (in 100 Monte Carlo runs) in the processing of the first measurement. As a typical case, the

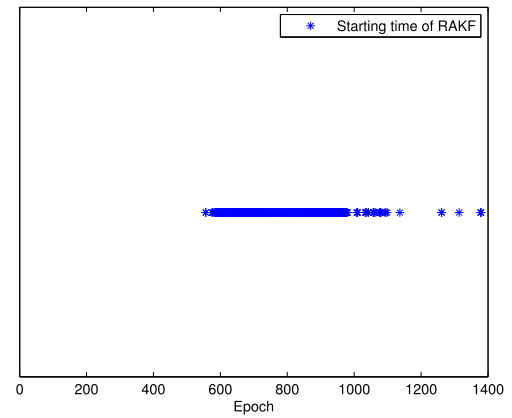


Fig. 5. Starting time of RAKF in the proposed method.

Table 1

Average RMSEs of position estimation over 100 Monte Carlo runs with $T_s = 0.1$ s.

Method	Proposed	NAR	RAKF	RAE	ARUF
Straight	4.03	4.17	6.15	4.38	4.54
Circular	3.25	8.15	9.17	17.05	10.23
All	4.90	6.00	6.55	8.28	6.36

Table 2

Average RMSEs of position estimation over 100 Monte Carlo runs with $T_s = 0.01$ s.

Method	Proposed	NAR	RAKF	RAE	ARUF
Straight	2.52	2.66	4.76	2.54	2.59
Circular	1.54	1.60	1.60	1.59	1.60
All	2.46	2.52	3.90	2.50	2.51

algorithm reduces to EKF in the subsequent updates. Therefore, it has no advantage over the proposed algorithm on adapting R . That is the reason for its worse performance than the proposed algorithm.

Tables 1 and 2 list the root mean square errors (RMSEs) of all the approaches over 100 Monte Carlo runs during different motions and the whole steady process with $T_s = 0.1$ s and $T_s = 0.01$ s, respectively.

In order to check the feasibility of the proposed algorithm, larger Q and T_s are implemented. The proposed algorithm also outperforms other algorithms with $Q = 10Q$ and $T_s = 0.5$ s, but the estimation accuracy declines because of larger time step.

From the simulation results, one can find that the performance of these filters is consistent with the theoretical analysis that has been stated in Sections 2–4.

6. Conclusion

This paper has documented a redundant measurement-based adaptive Kalman filter (RAKF). A PSOMD algorithm has been derived to estimate R in the system with single measurement by establishing a pseudo measurement. To complete the adaptive filtering, we have combined PSOMD with RAKF and proposed a rule for detecting estimation bias. If the bias occurs, the RAKF is implemented as compensatory actions. Simulation of a target tracking problem has been conducted to demonstrate the efficacy of the proposed approach, the results show satisfactory performance of the proposed algorithm. Although the estimated R can be impervious to the state estimation error when T_s is small, it is still affected by the process noise covariance Q . Is there any possibility that we could find a way to make the estimated R equally unaffected by Q ? This deserves further research.

Acknowledgments

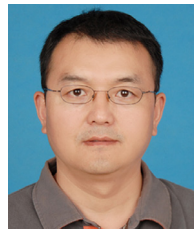
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